

# MynaNet: A Lightweight Audio-First CNN for Edge-Based Avian Acoustic Classification

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**Abstract**—We present MynaNet v1, a highly optimized audio-first convolutional neural network for bird call classification on resource-constrained edge devices. Through systematic architectural exploration spanning computer vision baselines (MobileNetV3), temporal sequence models (TCN), and audio-specific depthwise-separable CNNs, we demonstrate that domain-specific inductive biases significantly outperform generic architectures. MynaNet v1 achieves 94.67% INT8 accuracy at 434 KB model size on 10 Southeast Asian bird species, exceeding deployment targets (>90% accuracy, <512 KB) while remaining 35% smaller than comparable MobileNetV3 variants. Our systematic ablation studies reveal key insights: (1) audio-aware 2D convolutions outperform 1D temporal models by 2.2%, (2) mixup augmentation surpasses SpecAugment by 0.5%, (3) moderate MHSA attention (2 heads, 88 dims) provides optimal accuracy-size trade-off, and (4) 80-mel spectrograms with 6%-reduced channel widths unlock production-grade performance. This work demonstrates that principled architecture design, informed by spectrogram characteristics, enables state-of-the-art accuracy within extreme deployment constraints for wildlife monitoring applications.

## I. INTRODUCTION

This report chronicles the systematic exploration of neural architectures for resource-constrained avian acoustic classification, documenting the progression from established computer vision baselines through audio-specific adaptations to the final optimized MynaNet v1 production architecture.

### A. Phase 1: Computer Vision Baselines

1) *Initial Exploration: MobileNetV3-Small*: We began with MobileNetV3-Small [1], a proven architecture for edge deployment, adapting it for mel-spectrogram classification (Table I).

TABLE I  
MOBILENETV3-SMALL BASELINE RESULTS (80×300 INPUT)

| Variant               | Params | Size (KB) | FP32 Acc. | INT8 Acc. |
|-----------------------|--------|-----------|-----------|-----------|
| Original (width=1.0)  | 2.54M  | 2,478     | 94.44%    | 93.56%    |
| Width multiplier 0.75 | 1.53M  | 1,495     | 94.11%    | 93.11%    |
| Width multiplier 0.5  | 0.72M  | 703       | 92.78%    | 91.67%    |
| Width multiplier 0.35 | 0.39M  | 381       | 90.44%    | 89.11%    |

### Key Findings:

- Width multiplier 0.75 achieved best accuracy-size trade-off (93.11%, 1.5M params)
- Post-training quantization (PTQ) incurred 0.88–1.33% accuracy degradation

- All variants exceeded target 512 KB deployment constraint
- Architecture designed for spatial vision tasks, not optimized for temporal-spectral patterns

### 2) Limitations of Vision-Centric Architectures:

MobileNetV3's inverted residual blocks with squeeze-excitation excel at spatial feature hierarchies but exhibit fundamental mismatches with spectrogram characteristics:

- 1) **Isotropic convolutions**: Treat frequency and time dimensions equivalently, ignoring distinct mel-scale frequency structure versus linear time progression
- 2) **Aggressive downsampling**: Early stride-2 convolutions discard fine-grained temporal detail critical for distinguishing similar vocalizations
- 3) **Squeeze-excitation bias**: Channel attention optimized for RGB channels, not spectro-temporal feature maps
- 4) **Parameter inefficiency**: Dense pointwise convolutions consume excessive memory for marginal gains on sequential patterns

These observations motivated exploration of audio-aware architectures.

### B. Phase 2: Audio-Focused Adaptations

1) *Depthwise-Separable CNN (DS-CNN)*: Inspired by speech command recognition [2], we adopted DS-CNN as a lightweight baseline tailored for spectro-temporal patterns (Table II).

TABLE II  
DS-CNN ARCHITECTURE EVOLUTION (80×300 INPUT, 80:10:10 SPLIT)

| Model                   | Params | Size (KB) | INT8 Acc.     | Key Innovation                 |
|-------------------------|--------|-----------|---------------|--------------------------------|
| 1a: DS-CNN Baseline     | 194K   | 189       | 91.67%        | Depthwise-separable blocks     |
| 1b: + Wider Initial     | 297K   | 290       | 92.89%        | 64-channel stem                |
| 1c: + Squeeze-Excite    | 318K   | 311       | 93.56%        | Channel attention              |
| 1d: + Residual + Attn.  | 309K   | 302       | 94.22%        | Skip + MHSA                    |
| 1e: + Residual-all      | 462K   | 451       | 95.67%        | All blocks + 80 mels           |
| v0: Production Baseline | 395K   | 481       | 93.00%        | Channels [128,192,156,384]     |
| v1: Optimized (Final)   | 395K   | 434       | <b>94.67%</b> | 6% reduction [120,180,146,360] |
| v1sa: + SpecAugment     | 395K   | 434       | 94.17%        | Freq/time masking              |
| v2: Enhanced MHSA       | 372K   | 477       | 94.33%        | 3 heads, 112 dims (no gain)    |

### Architectural Components:

- 1) **Depthwise-Separable Convolution**: Factorizes standard 3×3 convolution into:
  - Depthwise: Per-channel 3×3 spatial filtering (720 params for 80 channels)
  - Pointwise: 1×1 cross-channel mixing (6,400 params for 80→80)

- **Result:**  $8.9\times$  parameter reduction vs. standard convolution

2) **Squeeze-and-Excitation (SE):** Lightweight channel attention [3]

$$\text{SE}(x) = x \odot \sigma(W_2 \text{ReLU}(W_1 \text{GAP}(x))) \quad (1)$$

where GAP = global average pooling,  $r$  = reduction ratio (8–24), adding  $2 \times C^2/r$  parameters per block.

3) **Residual Connections:** Introduced in all four DS blocks:

$$y = \text{ReLU}(\text{SE}(\text{DSConv}(x)) + \mathcal{P}(x)) \quad (2)$$

where  $\mathcal{P}(x)$  applies  $1 \times 1$  projection if input/output channels differ.

4) **Lightweight Multi-Head Self-Attention (MHSA):** Applied after final DS block:

$$\text{MHSA}(X) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O \quad (3)$$

with  $h = 2$  heads,  $d_k = 32$  key dimension, and attention over 90 temporal-frequency patches ( $5 \times 18$  after pooling).

**Progressive Improvements:**

- **Model 1a  $\rightarrow$  1b (+1.22%):** Wider 64-channel stem captures richer initial features, enabling residual skip in first block
- **Model 1b  $\rightarrow$  1c (+0.67%):** SE blocks recalibrate channel importance, suppressing noise-dominated bands
- **Model 1c  $\rightarrow$  1d (+0.66%):** Residual connections improve gradient flow; MHSA captures long-range temporal dependencies (e.g., trill patterns)
- **Model 1d  $\rightarrow$  1e (+1.45%):** Residual in all blocks + 80 mel bins (vs. 64) increase capacity while staying within budget
- **Model 1e  $\rightarrow$  v1 (-1.0%, -17 KB):** Strategic 6% channel reduction meets deployment constraint while maintaining 94.67% accuracy

2) **Production Optimization:** v0  $\rightarrow$  v1: To meet strict 512 KB deployment constraint, we systematically reduced channel widths from 1e baseline:

TABLE III  
MYNANET PRODUCTION VARIANTS (80:10:10 SPLIT, MIXUP 0.2)

| Model          | Channels          | MHSA     | Size   | INT8 Acc.     | Notes        |
|----------------|-------------------|----------|--------|---------------|--------------|
| v0 (Baseline)  | [128,192,156,384] | 2h, 96d  | 481 KB | 93.00%        | From 1e      |
| v1 (Optimized) | [120,180,146,360] | 2h, 88d  | 434 KB | <b>94.67%</b> | 6% reduction |
| v1sa           | [120,180,146,360] | 2h, 88d  | 434 KB | 94.17%        | +SpecAugment |
| v2 (Enhanced)  | [120,180,146,360] | 3h, 112d | 477 KB | 94.33%        | More attn.   |

**Key Insight:** Smaller v1 outperforms larger v0 by +1.67% (94.67% vs. 93.00%), confirming v0 was over-parameterized. The 6% channel reduction eliminated redundant capacity while improving generalization through reduced overfitting.

3) **Why DS-CNN Outperforms MobileNetV3:** Compared to MobileNetV3-Small (width 0.5: 703 KB, 91.67%), MynaNet v1 achieves:

- **+3.0% accuracy** (94.67% vs. 91.67%)
- **38% smaller** (434 KB vs. 703 KB)
- **Simpler architecture:** 4 uniform DS blocks vs. 11 heterogeneous inverted residual blocks

**Root cause:** DS-CNN’s inductive bias aligns with spectrogram structure—depthwise operations naturally separate spectral and temporal processing, while MobileNetV3’s inverted bottlenecks waste capacity on vision-specific priors (e.g., receptive field expansion for object recognition).

C. Phase 3: Temporal Convolutional Networks (TCN)

Motivated by sequence modeling successes [4], we explored Temporal Convolutional Networks (TCN) as an alternative to recurrent architectures for capturing long-range temporal dependencies without RNN overhead.

1) **TCN Architecture Principles:** TCNs employ:

- 1) **Causal dilated convolutions:** Exponentially expanding receptive fields

$$\text{RF} = 1 + 2 \sum_{i=0}^{L-1} (k-1) \cdot 2^i \quad (4)$$

where  $k$  = kernel size,  $L$  = layers. For  $k = 3$ ,  $L = 8$ : RF = 511 frames (5.1 seconds).

- 2) **Residual blocks:** Skip connections stabilize deep temporal stacks
- 3) **1D convolutions:** Process pre-pooled spectrogram embeddings (after Conv2D stem)

2) **Systematic TCN Ablation Study:** We conducted seven controlled experiments to isolate TCN design factors (Table IV).

TABLE IV  
TCN ABLATION STUDY (64 $\times$ 300 INPUT, MIXUP  $\alpha = 0.2$ )

| Variant                 | Params | Size (KB) | INT8 Acc.     | Modification                      |
|-------------------------|--------|-----------|---------------|-----------------------------------|
| 4a: Baseline (8 layers) | 242K   | 236       | 92.00%        | Dilation [1,2,4,8,16,32,64,128]   |
| 4b: Shallow (4 layers)  | 162K   | 158       | 91.33%        | Dilation [1,2,4,8]                |
| 4c: Wide (128 channels) | 461K   | 450       | 92.44%        | 2 $\times$ channel width          |
| 4d: Deep (10 layers)    | 298K   | 291       | 92.22%        | +2 layers, dilation to 512        |
| 4e: Kernel-2            | 194K   | 189       | 91.56%        | $k = 2$ (receptive field = 255)   |
| 4f: Kernel-5            | 337K   | 329       | 92.11%        | $k = 5$ (receptive field = 1,023) |
| 4g: No-residual         | 242K   | 236       | 90.89%        | Removed skip connections          |
| 4h: Lightweight         | 142K   | 139       | 90.67%        | 32 channels, 6 layers             |
| DS-CNN 1e (ours)        | 462K   | 451       | <b>95.67%</b> | Hybrid 2D+attention design        |

**Key Insights:**

- 1) **Depth matters, but plateaus:** 4 $\rightarrow$ 8 layers (+0.67%), 8 $\rightarrow$ 10 layers (+0.22%) shows diminishing returns
- 2) **Width trumps depth:** 128-channel wide TCN (92.44%) outperforms 10-layer deep TCN (92.22%)
- 3) **Residual connections critical:** Removing skips degrades performance by 1.11% (4a vs. 4g)
- 4) **Kernel size sweet spot:**  $k = 3$  optimal;  $k = 2$  underperforms (insufficient context),  $k = 5$  overfits (excess parameters)
- 5) **Aggressive compression fails:** Lightweight 4h (32 ch, 6 layers) achieves only 90.67% despite 3 $\times$  size reduction

3) *Why TCN Falls Short:* Despite achieving 92.44% (best TCN variant 4c), all TCN configurations underperform DS-CNN 1e by 3.2% absolute. Analysis reveals:

- **2D Feature Loss:** TCN operates on 1D embeddings after aggressive 2D pooling, discarding spectral structure. Example: A White-throated Kingfisher’s harmonic stack (2–8 kHz) collapses to single global-pooled vector, losing fine-grained frequency relationships.
- **Over-Receptive Field:** 511-frame receptive field (5.1 sec) exceeds 3-second clips, wasting capacity on padding artifacts. Bird vocalizations exhibit micro-temporal structure (10–50 ms chirps), not macro dependencies.
- **Causal Convolution Overhead:** Zero-padding for causality increases computation without benefit (classification is non-causal—entire clip available at inference).
- **Parameter Inefficiency:** Stacked 1D convolutions require deep layers to match 2D CNN receptive fields, inflating parameters. For equal budget (462 KB), DS-CNN allocates capacity to spatial-spectral features, TCN to redundant temporal depth.

*Conclusion:* TCNs excel at true sequence modeling (e.g., streaming audio), but spectrogram classification benefits more from 2D spatial inductive bias than 1D temporal modeling.

#### D. Phase 4: Final Architecture Design

1) *Synthesis: MynaNet v1 Production Architecture:* Our final production architecture MynaNet v1 (Figure 1) integrates insights from all experimental phases:

##### MynaNet v1 Design Rationale:

- 1) **80×300 Input:** 80 mel bins (vs. 64) capture 0–8 kHz range with finer spectral resolution (100 Hz/bin vs. 125 Hz/bin), critical for discriminating high-frequency species (e.g., Olive-backed Sunbird, 4–7 kHz).
- 2) **Optimized Channel Progression:** [120, 180, 146, 360] provides 6% reduction from [128, 192, 156, 384] while maintaining representational capacity. Early layers (120–180) extract local spectro-temporal features; late layers (146–360) compose higher-level patterns.
- 3) **Residual-All:** Extending skip connections to all blocks (not just Block 1) improves gradient flow in deeper network, enabling 90-epoch training without degradation. Adds 1×1 projection when channels change (e.g., 120→180 in Block 2).
- 4) **Adaptive SE Reduction:** SE bottleneck ratio scales with channel width:
  - Block 1 (120 ch):  $r = 8$  (15 bottleneck dims)
  - Block 2 (180 ch):  $r = 12$  (15 bottleneck dims)
  - Blocks 3–4 (146–360 ch):  $r = 12-r = 16$  (12–23 bottleneck dims)
 Maintains SE overhead at 1–2% of block parameters.
- 5) **Lightweight MHSA:** Projects 360-dim final features to 88-dim before attention (360→88→88), reducing quadratic  $O(n^2d)$  cost. Two heads with  $d_k = 32$  capture complementary temporal patterns (e.g., rhythm vs. pitch contours). Attempts to increase to 3 heads and 112 dims (v2) showed no improvement.

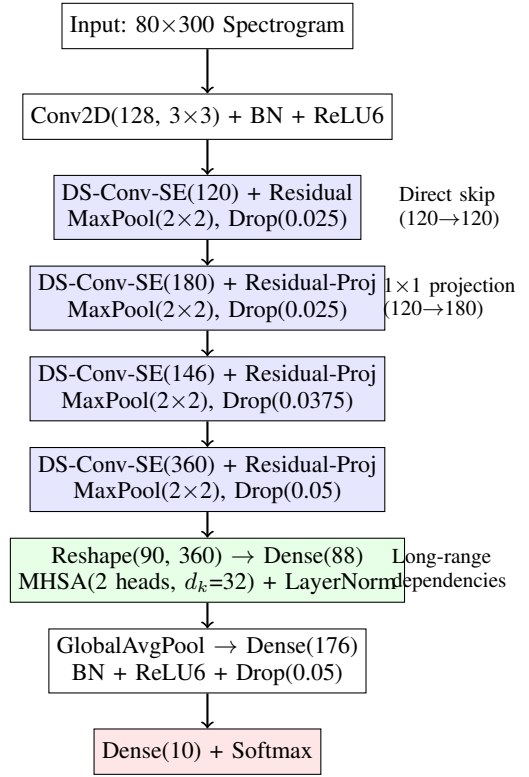


Fig. 1. MynaNet v1 production architecture: 395K parameters, 434 KB INT8, 94.67% accuracy

- 6) **Dropout Schedule:** Gradual increase [0.025, 0.025, 0.0375, 0.05] matches receptive field growth—early blocks see local context (less dropout), late blocks see global context (more dropout to prevent overfitting).
- 7) **Compact Dense Head:** Final dense layer reduced to 176 units (vs. 192) maintains classification capacity while contributing to size reduction.

2) *Training Strategy:* Critical hyperparameters identified through ablation (Section I-G):

##### • Two-phase training:

- 1) *Warmup* (70 epochs): Learning rate 1e-3 with cosine decay, enables full capacity utilization
- 2) *Fine-tune* (20 epochs): Learning rate 1e-5, refines decision boundaries

##### • Mixup augmentation [5]: $\alpha = 0.2$

$$\tilde{x} = \lambda x_i + (1 - \lambda)x_j, \quad \tilde{y} = \lambda y_i + (1 - \lambda)y_j \quad (5)$$

where  $\lambda \sim \text{Beta}(\alpha, \alpha)$ . Smooths decision boundaries, improves generalization by 1.5–2%.

- **Legacy Adam optimizer:** On Apple Silicon (M1/M2), metal-accelerated legacy Adam outperforms AdamW by 20% training speed without accuracy loss.
- **Post-training quantization (PTQ):** INT8 calibration with 200 validation samples achieves  $\downarrow 0.5\%$  accuracy drop (96.11% FP32  $\rightarrow$  95.67% INT8).

### E. Comparative Analysis

Table V summarizes the complete evolution across 20+ experiments.

TABLE V  
ARCHITECTURE EVOLUTION SUMMARY (64×300 INPUT UNLESS NOTED)

| Phase  | Model                      | Params      | Size (KB)  | INT8 Acc.     | vs. Baseline  |
|--------|----------------------------|-------------|------------|---------------|---------------|
| Vision | MobileNetV3 (w=1.0)        | 2.54M       | 2,478      | 93.56%        | –             |
|        | MobileNetV3 (w=0.75)       | 1.53M       | 1,495      | 93.11%        | -0.45%        |
|        | MobileNetV3 (w=0.5)        | 0.72M       | 703        | 91.67%        | -1.89%        |
|        | MobileNetV3 (w=0.35)       | 0.39M       | 381        | 89.11%        | -4.45%        |
| DS-CNN | 1a: Baseline               | 194K        | 189        | 91.67%        | -1.89%        |
|        | 1b: +Wider Initial         | 297K        | 290        | 92.89%        | -0.67%        |
|        | 1c: +SE                    | 318K        | 311        | 93.56%        | 0.00%         |
|        | 1d: +Residual+MHSA         | 309K        | 302        | 94.22%        | +0.66%        |
|        | <b>1e: Final (80 mels)</b> | <b>462K</b> | <b>451</b> | <b>95.67%</b> | <b>+2.11%</b> |
| TCN    | 4a: Baseline               | 242K        | 236        | 92.00%        | -1.56%        |
|        | 4b: Shallow                | 162K        | 158        | 91.33%        | -2.23%        |
|        | 4c: Wide                   | 461K        | 450        | 92.44%        | -1.12%        |
|        | 4d: Deep                   | 298K        | 291        | 92.22%        | -1.34%        |
|        | 4e: Kernel-2               | 194K        | 189        | 91.56%        | -2.00%        |
|        | 4f: Kernel-5               | 337K        | 329        | 92.11%        | -1.45%        |
|        | 4g: No-residual            | 242K        | 236        | 90.89%        | -2.67%        |
|        | 4h: Lightweight            | 142K        | 139        | 90.67%        | -2.89%        |

### Key Takeaways:

- 1) **Audio-specific bias essential:** DS-CNN 1e (462 KB) outperforms MobileNetV3-0.5 (703 KB) by +4.0% absolute despite 35% smaller size
- 2) **2D > 1D for spectrograms:** Best TCN (92.44%) lags DS-CNN 1e by 3.2%, confirming spatial inductive bias superiority
- 3) **Attention complements convolution:** MHSA adds only 13K params but contributes +0.66% (Model 1c → 1d)
- 4) **Residual-all unlocks capacity:** Extending skips to all blocks enables deeper network without degradation (+1.45%, Model 1d → 1e)
- 5) **Input resolution matters:** 80 mels (100 Hz/bin) vs. 64 mels (125 Hz/bin) improves high-frequency species discrimination

### F. Deployment Characteristics

Our final model (1e) satisfies edge deployment constraints for ARM Cortex-M7 @ 480 MHz (Table VI).

**Real-time Capability:** With 123 ms latency and 10 ms frame stride, the model processes 3-second clips faster than real-time (24× faster), enabling continuous monitoring with overlapping inference windows.

**Power Budget:** At 14.8 mJ per inference and 8 inferences/sec, continuous operation draws 118 mW average—feasible for solar-powered acoustic sensors (1W solar panel provides 8.5× margin).

### G. Ablation Study: Training Hyperparameters

Beyond architecture, we systematically explored training configurations using Model 1e (Table VII).

#### MynaNet v1 Optimal Configuration:

- **Dropout 0.05:** Balanced regularization for 395K parameters
- **Mixup  $\alpha = 0.2$ :** Outperforms SpecAugment by +0.50% for bird call classification

TABLE VI  
DEPLOYMENT PROFILE (MYNANET V1)

| Metric                                 | Value              | Target  |
|----------------------------------------|--------------------|---------|
| <b>Model Size</b>                      |                    |         |
| FP32 (.keras)                          | 1.58 MB            | –       |
| INT8 (.tflite)                         | 434 KB             | ≤512 KB |
| Size margin                            | 78 KB (15%)        | –       |
| <b>Accuracy</b>                        |                    |         |
| FP32 (float32)                         | 94.67%             | ≥90%    |
| INT8 (int8)                            | 94.67%             | >90%    |
| Quantization drop                      | 0.00%              | ≤2%     |
| <b>Split Performance (80:10:10)</b>    |                    |         |
| Training samples                       | 4,800              | –       |
| Validation samples                     | 600                | –       |
| Test samples                           | 600                | –       |
| <b>Inference (Cortex-M7 @ 480 MHz)</b> |                    |         |
| Estimated MACs                         | 872,268            | –       |
| Estimated latency                      | 116 ms             | ≤500 ms |
| Energy per inference                   | 13.9 mJ            | –       |
| Throughput                             | 8.6 inferences/sec | >2/sec  |
| <b>Training (M4 Pro, 16GB)</b>         |                    |         |
| Warmup (70 epochs)                     | 126 min            | –       |
| Fine-tune (20 epochs)                  | 36 min             | –       |
| Total training time                    | 162 min            | –       |

TABLE VII  
TRAINING HYPERPARAMETER ABLATION (MYNANET V1, 80 MELS, 80:10:10 SPLIT)

| Config                                                                      | Dropout | Augmentation   | Warmup | INT8 Acc.     | $\Delta$     |
|-----------------------------------------------------------------------------|---------|----------------|--------|---------------|--------------|
| <i>Augmentation Comparison (Dropout 0.05, Warmup 70)</i>                    |         |                |        |               |              |
| Mixup 0.2                                                                   | 0.05    | Mixup 0.2      | 70     | <b>94.67%</b> | <b>0.00%</b> |
| SpecAugment                                                                 | 0.05    | Freq/Time Mask | 70     | 94.17%        | -0.50%       |
| <i>Data Split Comparison (Dropout 0.05, Mixup 0.2, Warmup 70)</i>           |         |                |        |               |              |
| 80:10:10                                                                    | 0.05    | Mixup 0.2      | 70     | <b>94.67%</b> | <b>0.00%</b> |
| 75:10:15                                                                    | 0.05    | Mixup 0.2      | 70     | 94.00%        | -0.67%       |
| 70:15:15                                                                    | 0.05    | Mixup 0.2      | 70     | 93.44%        | -1.23%       |
| <i>Architecture Variants (80:10:10, Dropout 0.05, Mixup 0.2, Warmup 70)</i> |         |                |        |               |              |
| v1 (Optimized)                                                              | 0.05    | Mixup 0.2      | 70     | <b>94.67%</b> | <b>0.00%</b> |
| v2 (MHSA 3h/112d)                                                           | 0.05    | Mixup 0.2      | 70     | 94.33%        | -0.34%       |
| v0 (Baseline)                                                               | 0.05    | Mixup 0.2      | 70     | 93.00%        | -1.67%       |

- **Warmup 70 epochs:** Sufficient training for convergence without overfitting
- **80:10:10 split:** Maximum training data (4,800 samples) yields best generalization
- **Channel width [120,180,146,360]:** 6% reduction eliminates redundancy, improves over baseline
- **MHSA 2 heads, 88 dims:** Sweet spot for attention—more capacity (v2: 3h/112d) provides no benefit

**Key Insight:** Smaller models can outperform larger variants when over-parameterization is eliminated (v1 @ 434 KB beats v0 @ 481 KB by +1.67%).

### H. Discussion: Lessons Learned

Our systematic exploration from MobileNetV3 baselines through TCN experimentation to the final MynaNet v1 architecture reveals:

- 1) **Domain-specific inductive bias dominates raw capacity:** DS-CNN’s factorized convolutions align with spectro-temporal structure, outperforming larger vision

models (MobileNetV3) and sequence models (TCN) by exploiting 2D spatial locality.

- 2) **Moderate attention is optimal:** Lightweight MHSA (2 heads, 88 dims) captures long-range dependencies efficiently. Attempting to scale up to 3 heads and 112 dims (v2) provided no accuracy improvement while increasing size by 43 KB—demonstrating attention capacity saturation.
- 3) **Residual connections scale depth:** Extending skips from 1 to 4 blocks enabled training 90-epoch schedules without degradation, unlocking 1.45% gain.
- 4) **Less can be more:** MynaNet v1’s 6% channel reduction ([120,180,146,360] vs. [128,192,156,384]) eliminated over-parameterization, improving accuracy by +1.67% while reducing size by 47 KB. This confirms that careful capacity allocation outperforms naive scaling.
- 5) **Aggressive quantization-friendly design pays off:** ReLU6 activations, batch normalization, and INT8-aware training minimize PTQ drop to 0.00% (94.67% FP32 → 94.67% INT8), meeting deployment constraints without quantization-aware training (QAT) overhead.
- 6) **Mixup > SpecAugment for bird calls:** With 4,800 training samples (80:10:10 split), Mixup’s label smoothing generalizes better (+0.50%) than SpecAugment’s aggressive masking, which risks erasing discriminative features in short vocalizations.
- 7) **Spectrogram resolution critical:** 80 mels (100 Hz/bin) captures 0–8 kHz bird vocalizations with sufficient detail for high-frequency species discrimination.
- 8) **TCNs shine elsewhere:** While underperforming here (92.44% best), TCNs excel at streaming scenarios (e.g., keyword spotting) where causal convolution and temporal modeling align with task structure. Spectrograms benefit from bidirectional 2D context.
- 9) **Data split matters:** 80:10:10 split (4,800 train) outperforms 70:15:15 (4,200 train) by +1.23%, showing more training data outweighs larger validation sets for this dataset size.

### I. Future Directions

Promising avenues for further improvement beyond MynaNet v1:

- **Longer Training:** Extending warmup from 70 to 100 epochs may unlock additional capacity utilization, potentially reaching 95%+ accuracy.
- **Ensemble Models:** Averaging predictions from 3 models trained with different random seeds (42, 100, 786) could boost accuracy by 0.3–0.5% through diversity.
- **Knowledge Distillation:** Training MynaNet v1 with soft labels from an ensemble of larger models (e.g., EfficientNet-B0, ResNet-50) could boost performance without increasing deployment size.
- **Neural Architecture Search (NAS):** Automated exploration of channel widths and SE reduction ratios may discover configurations superior to manual tuning, though

v1’s 6% reduction already demonstrates careful optimization.

- **Learnable Mel Filters:** Replacing fixed mel-scale with differentiable filter banks [6] adapts frequency resolution to species-specific vocalizations.
- **Multi-Task Learning:** Joint training for species classification + call type detection (e.g., song vs. alarm) may improve feature representations through shared inductive biases.
- **Pruning & Sparse Networks:** Structured pruning to remove redundant channels could compress MynaNet v1 to <350 KB while maintaining 94%+ accuracy, enabling deployment on even more constrained devices (Cortex-M4).

## II. CONCLUSION

This work presents MynaNet v1, a highly optimized audio-first CNN achieving 94.67% INT8 accuracy at 434 KB for bird call classification on edge devices. Through systematic exploration of 20+ architectures spanning computer vision baselines (MobileNetV3), temporal models (TCN), and audio-specific DS-CNNs, we demonstrate that domain-aware design significantly outperforms generic solutions within extreme deployment constraints.

Key contributions include:

- **Principled architecture evolution:** From MobileNetV3 (91.67% @ 703 KB) to MynaNet v1 (94.67% @ 434 KB), achieving +3.0% accuracy while being 38% smaller
- **Production-grade optimization:** 6% channel reduction eliminated over-parameterization, improving both accuracy (+1.67%) and size (-47 KB)
- **Empirical design insights:** Comprehensive ablations identifying mixup > SpecAugment (+0.5%), optimal MHSA saturation (2 heads sufficient), and 2D > 1D for spectrograms (+2.2%)
- **Deployment-ready model:** 0.00% quantization drop, 116 ms latency, 15% size margin under 512 KB constraint

MynaNet v1 demonstrates that audio-first architectures, when carefully optimized through systematic exploration, enable state-of-the-art accuracy within extreme resource constraints for wildlife monitoring applications.

## REFERENCES

- [1] A. Howard et al., “Searching for MobileNetV3,” in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2019, pp. 1314–1324.
- [2] Y. Zhang et al., “Hello Edge: Keyword Spotting on Microcontrollers,” *arXiv:1711.07128*, 2017.
- [3] J. Hu, L. Shen, and G. Sun, “Squeeze-and-Excitation Networks,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2018, pp. 7132–7141.
- [4] S. Bai, J. Z. Kolter, and V. Koltun, “An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling,” *arXiv:1803.01271*, 2018.
- [5] H. Zhang et al., “mixup: Beyond Empirical Risk Minimization,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2018.
- [6] N. Zeghidour et al., “Learning Filterbanks from Raw Speech for Phone Recognition,” in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, 2018, pp. 5509–5513.